Diket: $a = 3$, $b = 2$, $n = 32$

A. Berdasarkan $N = 32$

$A = \{32, 34, 36, 39, 42, 45, 47\}$

$B = \{35, 37, 39, 42, 44, 50\}$

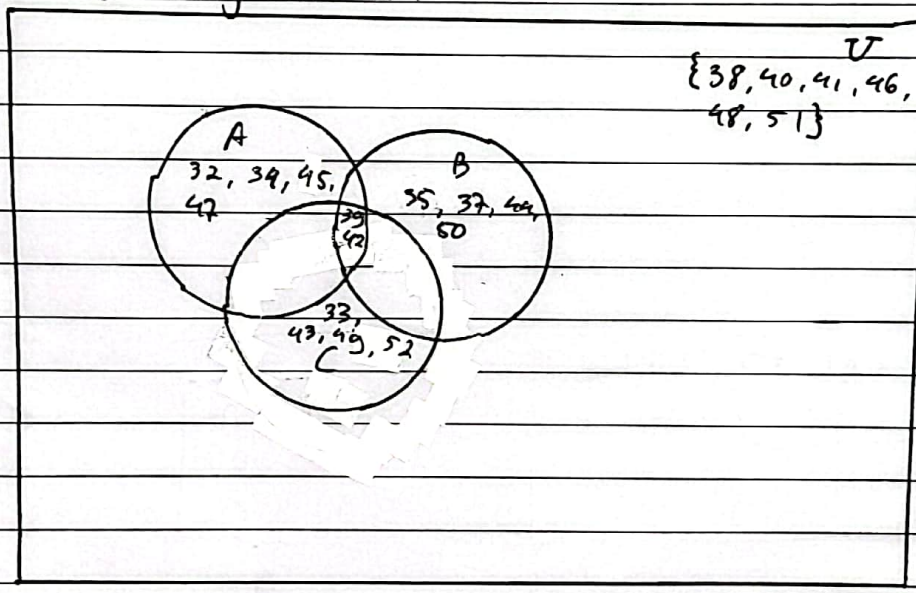
$C = \{33, 39, 42, 43, 49, 52\}$

$U = \{32, 33, \ldots, 52\}$ (total 21 elemen)

Irisan dan gabungan:

- $A \cap B \cap C = \{39, 42\}$ (elemen yg ada di ketiga himpunan)
- $A \cup B \cup C = \{32, 33, 34, \ldots, 52\}$

Gambar diagram venn:



Fungsi Invers dan Volume Pembelian

Invers $f^{-1}(x)$:

$y = 6x + 5 \rightarrow x = \frac{y-5}{6}$,

maka $f^{-1}(x) = \frac{x-5}{6}$.

Mencari volume x:

Pendapatan $y = (6 \cdot 100 + 3 + 2)$

$= 605$

$x = \frac{605-5}{6} = \frac{600}{6} = 100$

Kardinalitas:

1. $|A \cup B \cup C| = 15$
2. $|A \cap B \cap C| = 2$
3. $|(A \cup B) \cap C'|$: $A \cup B = \{32, 34, 35, 36, 37, 38, 42, 44, 45, 47, 50\}$

   $C' = U - C = \{32, 34, 35, 36, 37, 40, 41, 44, 45, 46, 47, 48, 50, 51\}$

   Irisannya $= \{32, 34, 35, 36, 37, 44, 45, 47, 50\}$, maka $|(A \cup B) \cap C'| = 9$

   $|A \oplus B|$ (symmetric diffrence):

   - $(A \cup B) - (A \cap B) = \{32, 34, 35, 36, 37, 44, 45, 47, 50\}$, maka $|A \oplus B| = 9$

Komposisi fungsi

Diket: $f(x) = 6x + 5$ dan $g(x) = \frac{x^2}{3} + 3$

Bukti komunatif ($x = b + 2 = 4$):

- $(f \circ g)(4) = 2(16) + 23 = 32 + 23 = 55$
- $(g \circ f)(4) = 12(16) + 20(4) + 11 = 33 = 192 + 80 + 11 = 33$
  $= 283 = 33$
- Kesimpulan: $(f \circ g)(x) \neq (g \circ f)(x)$, maka tdk komunatif.

- $(f \circ g)(x) = f(g(x)) = 6\left(\frac{x^2}{3} + 3\right) + 5 = 2x^2 + 18 + 5 = 23$
- $(g \circ f)(x) = g(f(x)) = \frac{(6x+5)^2}{3} + 3 = \frac{36x^2 + 60x + 25}{3} + 3 = 12x^2 + 20 + \frac{34}{3}$

Dipindai dengan CamScanner

Klasifikasi Fungsi

Fungsi $f(x) = 6x + 5$:

- Injektif? ya, krn setiap x yg berbeda menghasilkan f(x) yg berbeda ($6x_1 + 5 = 6x_2 + 5 \rightarrow x_1 = x_2$).
- Surjektif? ya, krn range fungsi adalah $(-\infty, \infty)$, mencakup seluruh kodominan R.
- Fungsi? ya, krn memenuhi ke 2nya

Fungsi $g(x) = \frac{x^2}{5} + 3$:

- Injektif? tdk, krn $g(x) = g(-x)$
- Surjektif? tdk, krn nilai g(x) selalu ≥ 3. kodominan bagian $(-\infty, 3)$ tdk memiliki pasangan di dominan.
- Fungsi? Tdk.

Dipindai dengan CamScanner